

0.1 RQ1-Specific Coding Codebook

Coding purpose. RQ1 examines how the included reviews characterize reinforcement-learning method families and directions, and which of them they favor for building control or future research. Two dimensions are coded: characterization and recommendation.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded family or direction within its own account of the literature. It does not measure the methodological quality, validity, or strength of the underlying primary-study evidence, and it does not measure how well a method performs in building control. A high score indicates that a review gives a topic a prominent and organizing place in its synthesis. Where a review reports that a family dominates the primary studies and builds its account around that observation, the prominence is recorded as such. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ1.

Score	Label	General interpretation
0	Not addressed	The family or direction is not identifiable in the review’s treatment of RL methods.
1	Mentioned	The family or direction, or its representative algorithms, is named or acknowledged, without substantive development of role, suitability, advantages, or limitations.
2	Discussed	The family or direction is substantively described, compared, or related to particular control settings, but does not organize a major part of the synthesis.
3	Structuring role	The family or direction organizes part of the review’s synthesis, through a taxonomy, repeated cross-study classification, dedicated section, major comparative table or figure, or recurring interpretive theme.

Two coding levels. RQ1 is coded at two levels. Unit-level scores record treatment intensity for each family or direction on each dimension, and populate the two cross-review heatmaps. Survey-level scores record the review’s overall treatment intensity on each dimension. The survey-level score equals the highest unit-level score recorded for that review on that dimension. A review reaches survey-level 3 when at least one unit reaches a structuring role, irrespective of how many units do so.

Coding units. The RQ1 axis covers method families and one method direction. Five algorithm families are coded across the corpus: value-based; policy-gradient / policy-based; actor-critic; model-based / hybrid; and multi-agent / distributed. One further unit records multi-objective RL (MORL), which several reviews use as an organizing frame instead of, or alongside, the family taxonomy. These six units are scored for every coded review on both dimensions, so a review that does not use a given framing receives 0 on that unit.

Three further labels appear in individual reviews’ own audit tables without being separate coding units. Deep RL is treated as a cross-cutting descriptor rather than a family, because it characterizes the function approximator rather than the learning rule; where a review uses it as a family label, that review’s audit table records the family to which it was mapped. Single-policy and multi-policy or adaptive strategies are recorded narratively for the reviews that organize their synthesis around policy strategy, and are not scored on the heatmaps.

Zero and NA. Two non-positive outcomes are distinguished and are not interchangeable.

Table 2: Normalization of review-specific labels to coding units. Mapping follows the classification each review itself uses, recorded in the corresponding audit table. Normalization changes labels only and leaves all scores unchanged.

Coding unit	Review-specific labels mapped to it
Value-based	Value-based; traditional RL
Policy-gradient / policy-based	Policy-gradient / policy-based; policy-search / policy-based; policy-based
Actor-critic	Actor-critic; actor-critic / advanced DRL; actor-critic / modern DRL
Model-based / hybrid	Model-based / hybrid; hybrid RL-MPC
Multi-agent / distributed	Multi-agent / distributed
MORL	MORL; multi-objective RL
Not coded as units	Deep RL, folded into the family it describes for that review; single-policy and multi-policy / adaptive, recorded narratively in the audit table

A score of 0 means the review does not address the unit. For recommendation, 0 also covers the case in which the review addresses the unit but declines to endorse it. Explicit non-endorsement is recorded as 0 and flagged in the audit note, so that it remains distinguishable from silence.

The scale does not separate a unit that a review considered and did not develop from one that fell outside the review’s declared scope. Both are recorded as 0. This is a deliberate simplification: it keeps the heatmap denominators uniform at the number of coded reviews, and it is consistent with the construct, which records the treatment a review gives a topic rather than the reasons for that treatment. The consequence is that a score of 0 should not be read as evidence that a review overlooked a topic it could have covered. Scope information for each review is reported separately in RQ3 and in the survey-level audit tables.

NA at the review level means unit-level coding is not an appropriate representation of the review as a whole. Such reviews are omitted from both heatmaps and retained in the qualitative synthesis.

Applicability. Unit-level coding applies when a review characterizes or favors RL methods in a form permitting cross-review comparison. Three reviews are not represented in the heatmaps. Sierla et al. (2022) organizes its synthesis around action formulation, actuator commands, set-points, and zone applicability rather than method comparison. Kannari et al. (2025) organizes its synthesis around real-building implementation, digital twins, safety, scalability, transparency, privacy, and governance. Both are marked NA by design and inform the qualitative RQ1 discussion. Togashi (2025) carries no RQ1 audit section and is therefore absent from the unit-level matrix, although it is coded on every other research question. The heatmap denominator is therefore 19 of 22, and every RQ1 count states its denominator.

Reviews that use a non-family organizing frame are retained rather than excluded, and are coded on the direction units alongside any family unit they do address. Huynh et al. (2025) organizes its synthesis around multi-objective formulations and policy strategies, and scores on MORL; its single-policy and multi-policy framings are recorded narratively in its audit table. It also treats classical single-objective methods under the label traditional RL, which is mapped to the value-based unit and scored 2 for characterization and 0 for recommendation, since the review presents these methods as a baseline it argues against rather than as an endorsed approach. The review’s zeros on the remaining family units record the absence of family-level treatment, not a judgment that it neglected those methods.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a unit in any of them supports a structuring score. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus and coding it would make reviews non-comparable. Coding reflects the review authors’ own account: a result present in a cited primary study is insufficient unless the review discusses, tabulates, or incorporates it. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Table 3: Operational criteria for unit-level characterization.

Score	Criterion
0	The unit is not identifiable in the review’s treatment of RL methods.
1	The unit or its representative algorithms are named or briefly acknowledged, without substantive development of role, prevalence, suitability, advantages, or limitations.
2	The unit is substantively explained, compared with other units, or related to particular building-control settings, but does not organize a major part of the review.
3	The unit is used as an organizing category in a taxonomy, repeated cross-study classification, comparative table or figure, dedicated synthesis, or recurring interpretation of the reviewed literature.

Unit-level operational criteria: characterization.

Table 4: Operational criteria for unit-level recommendation.

Score	Criterion
0	No identifiable endorsement. Description, prevalence, or historical dominance alone does not constitute recommendation. Explicit non-endorsement is also scored 0 and flagged in the audit note.
1	The unit is described as promising, potentially useful, or worth further investigation, but the endorsement is tentative or qualified by stated immaturity.
2	The review supports the unit under specified conditions, such as particular action spaces, building types, control objectives, scales, or deployment requirements.
3	The unit is positioned as a major preferred approach, a principal future direction, or a central solution to a recurring building-control problem.

Unit-level operational criteria: recommendation.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Table 5: Worked boundary examples for RQ1 coding.

Review	Unit and dimension	Score	Basis for the boundary
Wang and Hong (2020)	Model-based, characterization	1	The unit is acknowledged and compared conceptually with MPC-like approaches, but no substantive position is developed beyond noting that model learning is time-consuming and expertise-dependent. Boundary against 2.
Wang and Hong (2020)	Policy-gradient, characterization	2	Substantively discussed as a rising direction supporting transfer across buildings, but it does not organize part of the synthesis. Boundary against 3.
Shaqour and Hagishima (2022)	Actor-critic, characterization	3	Used as an organizing category across building-type comparison tables and recurring through the synthesis.
Han et al. (2021)	Actor-critic, recommendation	1	Presented as a distinct alternative for complex settings, with the review stating that more applications need to be developed. Tentative endorsement. Boundary against 2.
Han et al. (2021)	Policy-gradient, recommendation	2	Support is explicit but conditional on large MDPs and continuous states. Boundary against 1 and 3.